Lesson 14 Teacher's Key: Raising the Stakes

Unreal Engine (optional advanced track) - Instructor Reference Only

VIDEO GAME CREATION

What This Lesson Teaches

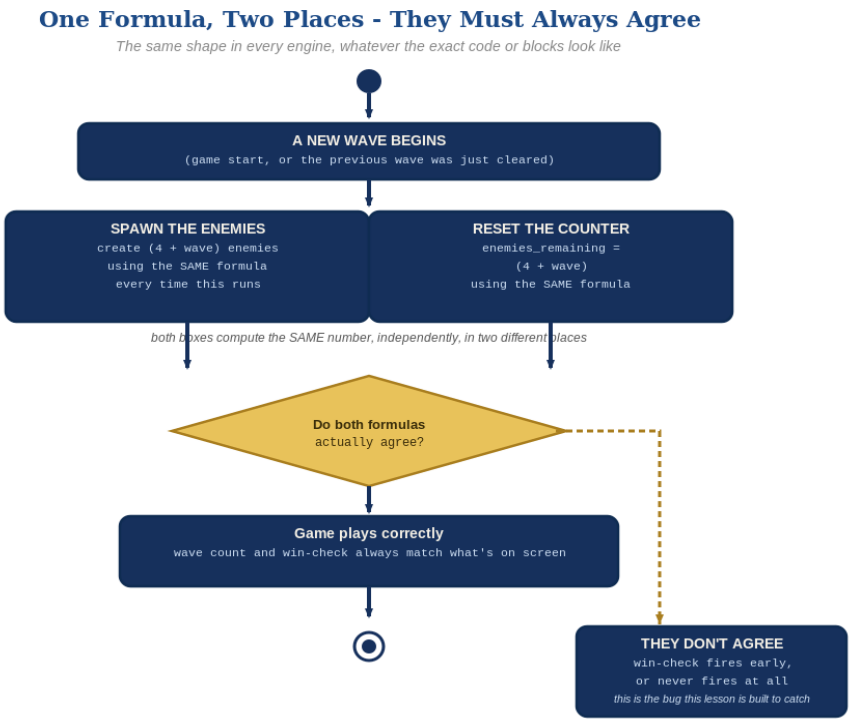
Same idea as every engine this week - replace hardcoded literals with wave-dependent formulas - expressed as Blueprint math nodes inside the SpawnWave Custom Event introduced in Lesson 11.

Level 1 - Full Worked Solution

- (4 + Wave) as the For Loop's end value and as EnemiesRemaining's reset value must be identical - both live inside the same SpawnWave graph, making it easy to verify they agree.
- No new variables are needed for Level 1 - this is purely a formula change inside a Custom Event that already exists.
- Wave 1 still spawns exactly 5 (4 + 1 = 5), so nothing about the game's opening feel changes - only later waves escalate.

Highlighted lines are new this week; everything else is code you already have.

```
Custom Event SpawnWave (revised) ->
  For Loop (0 to (4 + Wave) - 1) -> Spawn Actor from Class(BP_Enemy) ->
Bind Event to OnEnemyDestroyed / OnEnemyEscaped
  EnemiesRemaining = 4 + Wave
```



Whether it's two places (most engines) or the loop and the reset living in genuinely different scripts (Godot, GameMaker), the rule is the same: one formula, referenced everywhere.

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Whether the spawn loop and the reset live in two places or two genuinely different scripts, the rule is the same: one formula, referenced everywhere.

Why This Structure

- Computing (4 + Wave) once via a shared math node chain (rather than two separate calculations) keeps the For Loop and EnemiesRemaining tied to the exact same value - the same discipline as every other engine this week, expressed as Blueprint wiring instead of a stored variable.
- Setting EnemySpeed on each spawned BP_Enemy right after Spawn Actor from Class (rather than changing BP_Enemy's own default) keeps the scaling logic entirely inside SpawnWave - BP_Enemy's Event Tick movement stays exactly as simple as Lesson 7 left it.

Level 2 / Level 3 Talking Points

- Ask students to trace SpawnWave for wave 3 by hand: (4 + 3) = 7 - a good comprehension check for reading Blueprint math wiring.
- For students on the C++ track: point out that the identical (4 + Wave) formula, expressed as a plain arithmetic expression instead of Blueprint nodes, is conceptually the exact same idea - worth naming that parallel explicitly.

Level 2 - complete code:

Highlighted lines are the Level 2 additions on top of the Level 1 code above.

```
Custom Event SpawnWave (Level 2 addition, on top of the Level 1 code above
- speed set on each spawned instance):
```

```
  For Loop (0 to (4 + Wave) - 1) -> Spawn Actor from Class(BP_Enemy) ->
Bind Event to OnEnemyDestroyed / OnEnemyEscaped
  Set EnemySpeed (target = the just-spawned BP_Enemy) = 50.0 + (Wave - 1)
* 15.0
  EnemiesRemaining = 4 + Wave

C++ track (SpawnWave(), same addition):
newEnemy->EnemySpeed = 50.f + (Wave - 1) * 15.f;
```

Level 3 - no single combined code example:

This week's Level 3 is a "choose ONE" menu of five independent stretch features (destructible shields, a power-up, a bonus enemy, formation-marching movement, or a title/game-over screen) rather than the usual set of small, combinable stretch goals - so a single "all goals combined" code block doesn't fit Rule 14's usual model, and isn't provided here. Full worked solutions for all five options, across the active engines, are planned for the Lesson 15 wrap-up "Complete Codebase" send-home materials instead. Use the Talking Points above and the Handout's Level 3 menu to guide students toward whichever option they choose.

Common Student Mistakes

- Using (4 + Wave) as the For Loop's end value without the "- 1" adjustment For Loop nodes typically need (since For Loop's end value is inclusive) - resulting in one extra enemy per wave. Double-check against your specific For Loop node's exact semantics.
- Updating the For Loop's count but forgetting to update EnemiesRemaining (or vice versa).
- Setting EnemySpeed on BP_Enemy's class defaults instead of the just-spawned instance, which would affect ALL enemies at once rather than scaling per wave.